

B.Tech. Degree VIII Semester Examination April 2013

EC/EB/EI 804(C) MECHATRONICS
(2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

- I. (a) What are the different types of guide ways used in CNC machines? (15)
(b) Explain the advantages of NC over conventional machine tools. (10)
OR
- II. (a) Write short notes on (i) accuracy and (ii) productivity as applied to CNC machines. (10)
(b) What are the components of a CNC machine? Explain with the help of block diagrams/sketches. (15)
- III. (a) What are the components of the hydraulic control system used in a CNC machine tool? Explain with the help of a block diagram. (15)
(b) What is inductosyn? Explain its working principle with the help of a neat figure. (10)
OR
- IV. (a) Explain the working of AC and DC servomotors. (15)
(b) With the help of neat sketches (example) explain the linear and circular interpolations. (10)
- V. (a) Explain machine vision. Discuss the following three functional components of vision system: (10)
(i) Image acquisition
(ii) Processing
(iii) Output or display
(b) Explain what is automated storage and retrieval systems (ASRS). What are the different types of ASRS? What are the different elements in ASRS? Explain. (15)
OR
- VI. (a) What are the different tape address formats used in NC/CNC programming? Explain each with typical examples. (15)
(b) What is group technology? Explain. (10)
- VII. (a) Explain the different sensors used in robotics with examples. (15)
(b) Explain the terms repeatability, precision and accuracy related to robotics. (10)
OR
- VIII. (a) What are the different robot configurations? Explain each with neat figures. (15)
(b) Explain robot language structure. Also explain the generations of robot programming languages. (10)
